

Craft 2: Cut Square

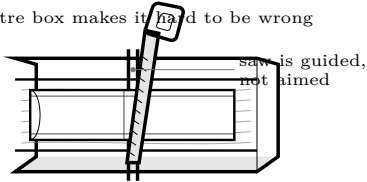
A joint can never be better than the cut that went into it

Shop skill — transferable to anything you ever build

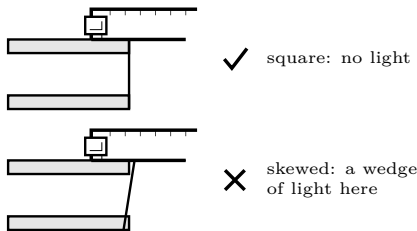
THE STANDARD

The cut face is **perpendicular to the pipe axis** within about one degree, and the whole face is flat. Held against a try square on a flat bench, you should not see light under the blade anywhere. On a 3 in pipe, one degree of skew is about 1.3 mm of gap on the far side — enough to matter, and easy to see once you know to look.

a mitre box makes it hard to be wrong



support the pipe on *both* sides of the cut



Why square matters

A socket joint is a precision taper. A skewed pipe end bottoms on one side while leaving a gap on the other, so the cement pools where it is not wanted and the weld area is smaller than it looks from outside. The joint appears finished and is not.

Use a guide

Freehand sawing a round object square is genuinely difficult and there is no reason to try. A mitre box or a tubing cutter turns a skill problem into an equipment problem.

Tubing cutter

Best result on pipe up to about 2 in — it cuts square by construction. Tighten a quarter turn per revolution, never more, or it ploughs and leaves a lip.

Fine teeth

On plastic, a fine-toothed saw cuts cleanly and a coarse one tears. Let the saw do the work; pushing hard makes the blade wander.

Support both sides

Pipe that drops as the cut finishes tears the last of the material and takes the face out of square in the final centimetre.

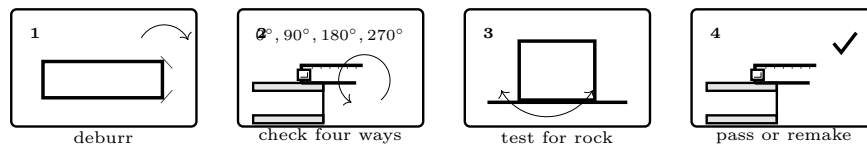
Cut on the waste side

Of the line, so the line survives on the piece you are keeping. *see the Craft 1: Measure & Mark sheet*

Method

1. Mark all the way round. A cut is only as square as its mark.
2. Set the pipe in the mitre box with the mark aligned to the slot, and hold it against the fence so it cannot roll.
3. Start the cut with the saw pulled backwards a couple of times to make a kerf. Do not start on the push.
4. Saw with long strokes and light pressure, letting the guide keep the angle. If thick pipe must be turned, stop, remove the saw, rotate the pipe, and secure it again before restarting. Never turn stock against a moving blade.
5. Support the offcut so it does not snap free.

Checking, and what to do if it is off



Test	What it tells you
Try square, light behind	The direct test. Rotate the square around the pipe to four positions — a cut can be square in one plane and skewed in the other.
Stand it on glass	A truly square end stands on a flat surface without rocking. Rock it gently; any wobble is skew.
Dry-fit and rotate	Push the pipe into its fitting and rotate it. If the gap at the shoulder changes as it turns, the end is not square.

Inspection decision

First remove only loose burrs; do not reshape the face. Ask: *Is there light at any of four square positions? Does the pipe rock on glass? Does the fitting shoulder gap change as the dry fit turns?*

Pass only if every answer is no and the finished length remains within 1 mm of the cut list. If any answer is yes, **re-cut from a new mark**. Do not sand the face square: it removes an unknown amount and can round the edge. If re-cutting would make the part undersize, **remake the part** from new stock. Keep the rejected piece only as a clearly marked practice coupon.

Final proof record

Finished length	_____ mm	Rocking on glass	no / yes
Light at 0°	no / yes	Light at 90°	no / yes
Light at 180°	no / yes	Light at 270°	no / yes
Rotating shoulder gap	no / yes	Cut-list tolerance	within 1 mm: no / yes

Circle the disposition: **PASS** / **RE-CUT** / **REMAKE**.

Inspector initials _____ Date _____

If not passed, record the failed test and the new cut mark or remake:

Only a recorded **PASS** proceeds to a joint.

The habit worth taking away

Every trade has a version of this: use a guide rather than your eye, and then check against a known-good reference rather than against your own judgement of how it looks. A try square is a piece of steel that is more perpendicular than you are. That is not an insult, it is the entire reason it exists.

Telos. Potato Launcher — Craft 2: Cut Square. Revised 2026-07-25. Project-created text and diagrams are licensed CC BY 4.0. Incorporated public-domain artwork remains public domain and is credited on the plate it appears with. See **LICENSE** and **CREDITS.md** in the source. Nothing here is professional advice; verify current regulations and follow the stated safety procedure.